

8 Arrow-Debreu time-zero trading (Chapter 8)

1. There are I agents, $i = 1, 2, \dots, I$. Endowment y_t^i , consumption c_t^i , utility

$$\sum_{t=0}^{\infty} \beta^t u(c_t^i)$$

2. Pareto Problem

$$\begin{aligned} \max \quad & \sum_i \lambda_i \left(\sum_{t=0}^{\infty} \beta^t u(c_t^i) \right) \\ \text{subject to} \quad & \sum_i c_t^i = \sum_i y_t^i \end{aligned}$$

3. Arrow-Debreu securities. Goods are indexed by calendar time.

4. **DEFINITION 7** A price system is $\{q_t\}_{t=0}^{\infty}$. An allocation is a list of consumptions $\{c_t^i\}_{t=0}^{\infty}$, one for each i . A competitive equilibrium is a feasible allocation and a price system such that, given the price system, the allocation solves each household's problem.

Proposition 2 The competitive equilibrium allocation is a function of the realized aggregate endowment and depends neither on the specific history leading up to that outcome nor on the realizations of individual endowments.

Example 8.6.3.

8.1 Add uncertainty

In each period, there is a realization of a stochastic event s_t . Key assumptions:

1. each household is small, hence is a price taker.
2. s_t is not realized until t . When it is realized, it is publicly observed by all households.
3. Households observe $y^i(s_t)$ for all i .
4. Households do not know the value of s_t in period 0, but they agree on its distribution.
5. There is perfect enforcement of contracts, i.e., households cannot default.

Any deviation from the above assumptions will significantly complicate the analysis.

Figure 8.3.1:

Repeat Pareto problem and Arrow-Debreu Equilibrium.

8.2 Asset Pricing

An asset entitles the owner to a stream of dividends: $\{d_t(s^t)\}_{t=0}^{\infty}$. Its price is

$$\sum_{t=0}^{\infty} \sum_{s^t} q_t(s^t) d_t(s^t)$$

EXAMPLE 15 In example 8.6.2., suppose a derivative asset promises to pay .05 units of consumption each period. What would the price of that asset be?

EXAMPLE 16 Consider the following version of the Lucas model of asset prices. There are I infinitely-lived people. Each household maximizes $E[\sum_{t=0}^{\infty} \beta^t \log(c_t)]$. There are two trees in the economy, a y -tree and a z -tree. Fruit of y -tree, y_t , is i.i.d. over time and is either I or $2I$. $\Pr(y_t = I) = 0.2$, $\Pr(y_t = 2I) = 0.8$. Fruit of z -tree, z_t is i.i.d. over time and is either I or $2I$. $\Pr(z_t = I) = 0.5$, $\Pr(z_t = 2I) = 0.5$. The two stochastic processes, $\{y_t\}$ and $\{z_t\}$ are independent.

Each household owns a share $\frac{1}{I}$ of y -tree and a share $\frac{1}{I}$ of z -tree. Compute the price of the y -tree if $y_0 = z_0 = I$.

$$\text{Solution: } c_0^i = 2 \text{ and } c_t^i = \begin{cases} 2, & \text{probability } 0.1, (y_t = I, z_t = I); \\ 3, & \text{probability } 0.1, (y_t = I, z_t = 2I); \\ 3, & \text{probability } 0.4, (y_t = 2I, z_t = I); \\ 4, & \text{probability } 0.4, (y_t = 2I, z_t = 2I); \end{cases}$$

Hence, for each t ,

$$\sum_{s^t} \pi(s^t) y(s^t) u'(c^i(s^t)) = 0.1 \frac{I}{2} + 0.1 \frac{I}{3} + 0.4 \frac{2I}{3} + 0.4 \frac{2I}{4} = 0.55I,$$

and the price of the y -tree is

$$q = \sum_{t=1}^{\infty} \beta^t 0.55I / u'(c_0) = \frac{\beta}{1-\beta} 1.1I$$

8.3 Sequential trading

This section describes an alternative market structure that achieves the same equilibrium allocation as Arrow-Debreu equilibrium.

8.3.1 Arrow securities

1. Market is reopened for trading each period.
2. At each date t , households trade a set of claims to one-period-ahead state-contingent consumption.

8.3.2 Sequential trading

Let $\tilde{a}_t^i(s^t)$ denote the claims to time t consumption, other than its endowment, that household i brings into time t in history s^t . $Q(s_{t+1}|s^t)$ gives the price of one unit of time- $(t+1)$ consumption, contingent on the realization s_{t+1} at $t+1$, when the history at t is s^t .

The household faces a sequence of budget constraints (at each history s^t)

$$c_t^i(s^t) + \sum_{s_{t+1}} Q(s_{t+1}|s^t) a^i(s^t, s_{t+1}) = y_t^i(s^t) + a^i(s^t),$$

where $a^i(s^0) = 0$ for all i . There is also a non-ponzi game condition that prevents the household from borrowing too much.

If $a^i(s^t) > 0$, household i is a lender in the security market. Otherwise he is a borrower. Total borrowing is equal to total lending, hence $\sum_i a^i(s^t) = 0$.

DEFINITION 8 A sequential-trading competitive equilibrium is an allocation $\{c^i(s^t), a^i(s^t)\}_{t=0}^{\infty}$, $i = 1, \dots, I$ and pricing kernels $\{Q(s_{t+1}|s^t)\}_{t=0}^{\infty}$ such that

1. for all i , given the pricing kernels, the allocation solves the household's problem;

2. $\sum_i a^i(s^t) = 0$ and $\sum_i c^i(s^t) = \sum_i y^i(s^t)$ for all s^t .

Review the FOC in Arrow-Debreu trading

$$\beta^t \pi(s^t) u'(c^i(s^t)) = \mu_i q(s^t),$$

where $\mu_i = u'(c^i(s^0))$. Hence it can be rewritten as

$$\beta^t \pi(s^t) \frac{u'(c^i(s^t))}{u'(c^i(s^0))} = q(s^t), \forall i = 1, 2, \dots, I.$$

FOC condition in Sequential Equilibrium.

$$L^i = \sum_t \sum_{s^t} \left(\beta^t \pi(s^t) u(c^i(s^t)) + \eta^i(s^t) \left(-c_t^i(s^t) - \sum_{s_{t+1}} Q(s_{t+1}|s^t) a^i(s^t, s_{t+1}) + y_t^i(s^t) + a^i(s^t) \right) \right)$$

$$\beta^t u'(c^i(s^t)) \pi(s^t) = \eta^i(s^t) \quad (8.8.5 \text{ a})$$

$$-\eta^i(s^t) Q(s_{t+1}|s^t) + \eta^i(s^{t+1}) = 0 \quad (8.8.5 \text{ b})$$

which simplifies to

$$Q(s_{t+1}|s^t) = \frac{\beta u'(c^i(s^{t+1})) \pi(s^{t+1})}{u'(c^i(s^t)) \pi(s^t)}, \quad (8.8.6)$$

8.3.3 Equivalence of allocations

THEOREM 12 *If $\{c^i(s^t)\}_{t=0}^\infty$ and $\{q(s^t)\}_{t=0}^\infty$ is a time-zero Arrow-Debreu equilibrium, then we can find prices of one-period-ahead arrow securities $Q(s_{t+1}|s^t)$ and asset holding $\{a^i(s^t)\}_{t=0}^\infty$ such that (c, Q, a) become a Sequential equilibrium.*

The definition of Q is given in (8.8.6).

$$Q(s_{t+1}|s^t) = \frac{q(s^{t+1})}{q(s^t)}.$$

How should we define $\{a^i(s^t)\}_{t=0}^\infty$? If they are defined appropriately, flow budget constraints will be satisfied. So we can infer $\{a^i(s^t)\}_{t=0}^\infty$ by consolidating budget constraints.

$$\begin{aligned}
q(s^t)c_t^i(s^t) + \sum_{s_{t+1}} q(s^{t+1})a^i(s^t, s_{t+1}) &= q(s^t)y_t^i(s^t) + q(s^t)a^i(s^t), \\
q(s^{t+1})c_{t+1}^i(s^{t+1}) + \sum_{s_{t+2}} q(s^{t+2})a^i(s^{t+1}, s_{t+2}) &= q(s^{t+1})y_{t+1}^i(s^{t+1}) + q(s^{t+1})a^i(s^{t+1}), \\
&\dots
\end{aligned}$$

Hence

$$\sum_{\tau=t}^{\infty} \sum_{s^\tau} q(s^\tau)c_\tau^i(s^\tau) - \sum_{\tau=t}^{\infty} \sum_{s^\tau} q(s^\tau)y_\tau^i(s^\tau) = q(s^t)a^i(s^t)$$

We can verify the flow budget constraint when we define $a^i(s^t)$ as above.

On the other hand, if we define

$$\begin{aligned}
q(s^0) &= 1 \\
q(s^{t+1}) &= q(s^t)Q(s_{t+1}|s^t),
\end{aligned}$$

then we have

THEOREM 13 *If $\{c^i(s^t), a^i(s^t)\}_{t=0}^\infty$ and $Q(s_{t+1}|s^t)$ is a Sequential-trading equilibrium, then we can find prices of history-contingent claims such that (c, q) become a time-zero Arrow-Debreu equilibrium.*

good $y_t^i = 1, 0, 0, 1, 0, 0, 1, \dots$. Consumer 2 is endowed with a stream of the consumption good $0, 1, 1, 0, 1, 1, 0, \dots$. Assume that there are complete markets with time-0 trading.

- a. Define a competitive equilibrium.
- b. Compute a competitive equilibrium.
- c. Suppose that one of the consumers markets a derivative asset that promises to pay .05 units of consumption each period. What would the price of that asset be?



Exercise 8.4 Consider a pure endowment economy with a single representative consumer; $\{c_t, d_t\}_{t=0}^{\infty}$ are the consumption and endowment processes, respectively. Feasible allocations satisfy

$$c_t \leq d_t.$$

The endowment process is described by¹⁶

$$d_{t+1} = \lambda_{t+1} d_t.$$

The growth rate λ_{t+1} is described by a two-state Markov process with transition probabilities

$$P_{ij} = \text{Prob}(\lambda_{t+1} = \bar{\lambda}_j | \lambda_t = \bar{\lambda}_i).$$

Assume that

$$P = \begin{bmatrix} .8 & .2 \\ .1 & .9 \end{bmatrix},$$

and that

$$\bar{\lambda} = \begin{bmatrix} .97 \\ 1.03 \end{bmatrix}.$$

In addition, $\lambda_0 = .97$ and $d_0 = 1$ are both known at date 0. The consumer has preferences over consumption ordered by

$$E_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\gamma}}{1-\gamma},$$

where E_0 is the mathematical expectation operator, conditioned on information known at time 0, $\gamma = 2, \beta = .95$.

¹⁶ See Mehra and Prescott (1985).

Part I

At time 0, after d_0 and λ_0 are known, there are complete markets in date- and history-contingent claims. The market prices are denominated in units of time-0 consumption goods.

a. Define a competitive equilibrium, being careful to specify all the objects composing an equilibrium.

b. Compute the equilibrium price of a claim to one unit of consumption at date 5, denominated in units of time-0 consumption, contingent on the following history of growth rates: $(\lambda_1, \lambda_2, \dots, \lambda_5) = (.97, .97, 1.03, .97, 1.03)$. Please give a numerical answer.

c. Compute the equilibrium price of a claim to one unit of consumption at date 5, denominated in units of time-0 consumption, contingent on the following history of growth rates: $(\lambda_1, \lambda_2, \dots, \lambda_5) = (1.03, 1.03, 1.03, 1.03, .97)$.

d. Give a formula for the price at time 0 of a claim on the entire endowment sequence.

e. Give a formula for the price at time 0 of a claim on consumption in period 5, contingent on the growth rate λ_5 being .97 (regardless of the intervening growth rates).

Part II

Now assume a different market structure. Assume that at each date $t \geq 0$ there is a complete set of one-period forward Arrow securities.

f. Define a (recursive) competitive equilibrium with Arrow securities, being careful to define all of the objects that compose such an equilibrium.

g. For the representative consumer in this economy, for each state compute the “natural debt limits” that constrain state-contingent borrowing.

h. Compute a competitive equilibrium with Arrow securities. In particular, compute both the pricing kernel and the allocation.

i. An entrepreneur enters this economy and proposes to issue a new security each period, namely, a risk-free two-period bond. Such a bond issued in period t promises to pay one unit of consumption at time $t+1$ for sure. Find the price of this new security in period t , contingent on λ_t .

the share price:

$$u'(y_t)R_t^{-1} = E_t\beta u'(y_{t+1}), \quad (13.5.1)$$

$$u'(y_t)p_t = E_t\beta(y_{t+1} + p_{t+1})u'(y_{t+1}). \quad (13.5.2)$$

13.6. Stock prices without bubbles

Using recursions on equation (13.5.2) and the law of iterated expectations, which states that $E_tE_{t+1}(\cdot) = E_t(\cdot)$, we arrive at the following expression for the equilibrium share price:

$$u'(y_t)p_t = E_t \sum_{j=1}^{\infty} \beta^j u'(y_{t+j})y_{t+j} + E_t \lim_{k \rightarrow \infty} \beta^k u'(y_{t+k})p_{t+k}. \quad (13.6.1)$$

Moreover, equilibrium share prices have to be consistent with market clearing; that is, agents must be willing to hold their endowments of trees forever. It follows immediately that the last term in equation (13.6.1) must be zero. Suppose to the contrary that the term is strictly positive. That is, the marginal utility gain of selling shares, $u'(y_t)p_t$, exceeds the marginal utility loss of holding the asset forever and consuming the future stream of dividends, $E_t \sum_{j=1}^{\infty} \beta^j u'(y_{t+j})y_{t+j}$. Thus, all agents would like to sell some of their shares and the price would be driven down. Analogously, if the last term in equation (13.6.1) were strictly negative, we would find that all agents would like to purchase more shares and the price would necessarily be driven up. We can therefore conclude that the equilibrium price must satisfy

$$p_t = E_t \sum_{j=1}^{\infty} \beta^j \frac{u'(y_{t+j})}{u'(y_t)} y_{t+j}, \quad (13.6.2)$$

which is a generalization of equation (13.3.4) in which the share price is an expected discounted stream of dividends but with time-varying and stochastic discount rates.

Note that asset bubbles could also have been ruled out by directly referring to transversality condition (13.2.7) and market clearing. In an equilibrium, the representative agent holds the per-capita outstanding number of shares.

(We have assumed one tree per person and one share per tree.) After dividing transversality condition (13.2.7) by this constant time-invariant number of shares and replacing c_{t+k} by equilibrium consumption y_{t+k} , we arrive at the implication that the last term in equation (13.6.1) must vanish.¹¹

Moreover, after invoking our assumption that the endowment follows a Markov process, it follows that the equilibrium price in equation (13.6.2) can be expressed as a function of the current state s_t ,

$$p_t = p(s_t). \quad (13.6.3)$$

13.7. Computing asset prices

We now turn to three examples in which it is easy to calculate an asset-pricing function by solving the expectational difference equation (13.5.2).

¹¹ Brock (1982) and Tirole (1982) use the transversality condition when proving that asset bubbles cannot exist in economies with a constant number of infinitely lived agents. However, Tirole (1985) shows that asset bubbles can exist in equilibria of overlapping generations models that are dynamically inefficient, that is, when the growth rate of the economy exceeds the equilibrium rate of return. O'Connell and Zeldes (1988) derive the same result for a dynamically inefficient economy with a growing number of infinitely lived agents. Abel, Mankiw, Summers, and Zeckhauser (1989) provide international evidence suggesting that dynamic inefficiency is not a problem in practice.

13.7.1. Example 1: Logarithmic preferences

Take the special case of equation (13.6.2) that emerges when $u(c_t) = \ln c_t$. Then equation (13.6.2) becomes

$$p_t = \frac{\beta}{1-\beta} y_t. \quad (13.7.1)$$

Equation (13.7.1) is our asset-pricing function. It maps the state of the economy at t , y_t , into the price of a Lucas tree at t .

13.7.2. Example 2: A finite-state version

Mehra and Prescott (1985) consider a discrete state version of Lucas's one-kind-of-tree model. Let dividends assume the n possible distinct values $[\sigma_1, \sigma_2, \dots, \sigma_n]$. Let dividends evolve through time according to a Markov chain, with

$$\text{prob}\{y_{t+1} = \sigma_l | y_t = \sigma_k\} = P_{kl} > 0.$$

The $(n \times n)$ matrix P with element P_{kl} is called a stochastic matrix. The matrix satisfies $\sum_{l=1}^n P_{kl} = 1$ for each k . Express equation (13.5.2) of Lucas's model as

$$p_t u'(y_t) = \beta E_t p_{t+1} u'(y_{t+1}) + \beta E_t y_{t+1} u'(y_{t+1}). \quad (13.7.2)$$

Express the price at t as a function of the state σ_k at t , $p_t = p(\sigma_k)$. Define $p_t u'(y_t) = p(\sigma_k) u'(\sigma_k) \equiv v_k$, $k = 1, \dots, n$. Also define $\alpha_k = \beta E_t y_{t+1} u'(y_{t+1}) = \beta \sum_{l=1}^n \sigma_l u'(\sigma_l) P_{kl}$. Then equation (13.7.2) can be expressed as

$$p(\sigma_k) u'(\sigma_k) = \beta \sum_{l=1}^n p(\sigma_l) u'(\sigma_l) P_{kl} + \beta \sum_{l=1}^n \sigma_l u'(\sigma_l) P_{kl}$$

or

$$v_k = \alpha_k + \beta \sum_{l=1}^n P_{kl} v_l,$$

or in matrix terms, $v = \alpha + \beta P v$, where v and α are column vectors. The equation can be represented as $(I - \beta P)v = \alpha$. This equation has a unique solution given by¹²

$$v = (I - \beta P)^{-1} \alpha. \quad (13.7.3)$$

¹² Uniqueness follows from the fact that, because P is a nonnegative matrix with row sums all equaling unity, the eigenvalue of maximum modulus P has

The price of the asset in state σ_k —call it p_k —can then be found from $p_k = v_k/[u'(\sigma_k)]$. Notice that equation (13.7.3) can be represented as

$$v = (I + \beta P + \beta^2 P^2 + \dots)\alpha$$

or

$$p(\sigma_k) = p_k = \sum_l (I + \beta P + \beta^2 P^2 + \dots)_{kl} \frac{\alpha_l}{u'(\sigma_k)},$$

where $(I + \beta P + \beta^2 P^2 + \dots)_{kl}$ is the (k, l) element of the matrix $(I + \beta P + \beta^2 P^2 + \dots)$. We ask the reader to interpret this formula in terms of a geometric sum of expected future variables.

13.7.3. Example 3: Asset pricing with growth

Let's price a Lucas tree in a pure endowment economy with $c_t = y_t$ and $y_{t+1} = \lambda_{t+1}y_t$, where λ_t is Markov with transition matrix P . Let p_t be the ex dividend price of the Lucas tree. Assume the CRRA utility $u(c) = c^{1-\gamma}/(1-\gamma)$. Evidently, the price of the Lucas tree satisfies

$$p_t = E_t \left[\beta \left(\frac{c_{t+1}}{c_t} \right)^{-\gamma} (p_{t+1} + y_{t+1}) \right].$$

Dividing both sides by y_t and rearranging gives

$$\frac{p_t}{y_t} = E_t \left[\beta (\lambda_{t+1})^{1-\gamma} \left(\frac{p_{t+1}}{y_{t+1}} + 1 \right) \right]$$

or

$$w_i = \beta \sum_j P_{ij} \lambda_j^{1-\gamma} (w_j + 1), \quad (13.7.4)$$

where w_i represents the price-dividend ration. Equation (13.7.4) was used by Mehra and Prescott (1985) to compute equilibrium prices.

modulus unity. The maximum eigenvalue of βP then has modulus β . (This point follows from Frobenius's theorem.) The implication is that $(I - \beta P)^{-1}$ exists and that the expansion $I + \beta P + \beta^2 P^2 + \dots$ converges and equals $(I - \beta P)^{-1}$.